

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0113

Fiesta 138 kV
May 24, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0113. The Interconnection Customer proposes a 173.6 MW wind facility interconnecting at the Fiesta 138 kV substation in the Virginia area. The request was received on March 09, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 93 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,348,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

LIGHTFOOT to TOANO 138kV line: 124.7% loading (234 MVA) following the outage of the MIRA VISTA to YARNEN 345kV; DFAX = 13.5%.

CHICO A SUBSTATION to YORK CANYON SUBSTATION 138kV: 108.2% loading (273 MVA) following the outage of the INDIAN FLAT to ACEC LEWISTON 138kV; DFAX = 5.4%.

The SYNAA HUGOTON to LIGHTFOOT 138kV is overloaded to 106.7% of its 307 MVA emergency rating for the loss of the MIRA VISTA to YARNEN 345kV L. The project contributes 13.5% DFAX (23.4 MW) to this overload.

The SYNATA HUGOTON to BATTLEFIELD 138kV is overloaded to 104.8% of its 339 MVA emergency rating for the loss of the MIRA VISTA to YARNEN 345kV. The project contributes 13.5% DFAX (23.4 MW) to this overload.

The HARDY REIT to BATTLEFIELD 138kV is overloaded to 103.1% of its 377 MVA emergency rating for the loss of the MIRA VISTA to YARNEN 345kV. The project contributes 13.5% DFAX (23.4 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LIGHTFOOT to TOANO 138kV line	Rebuild line with higher-capacity conductor	\$3,992,000	100% (\$3,992,000)
YORK CANYON to CHICO A 138kV	Reconductor line	\$2,661,000	100% (\$2,661,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SYNATA HUGOTON SUBSTATION to LIGHTFOOT 138kV	Reconductor line	\$1,867,000	100% (\$1,867,000)
SYNATA HUGOTON to BATTLEFIELD 138kV	Reconductor line	\$838,000	100% (\$838,000)
HARDY REIT to BATTLEFIELD 138kV	Reconductor line	\$6,990,000	100% (\$6,990,000)
	Total allocated to project		\$16,348,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.